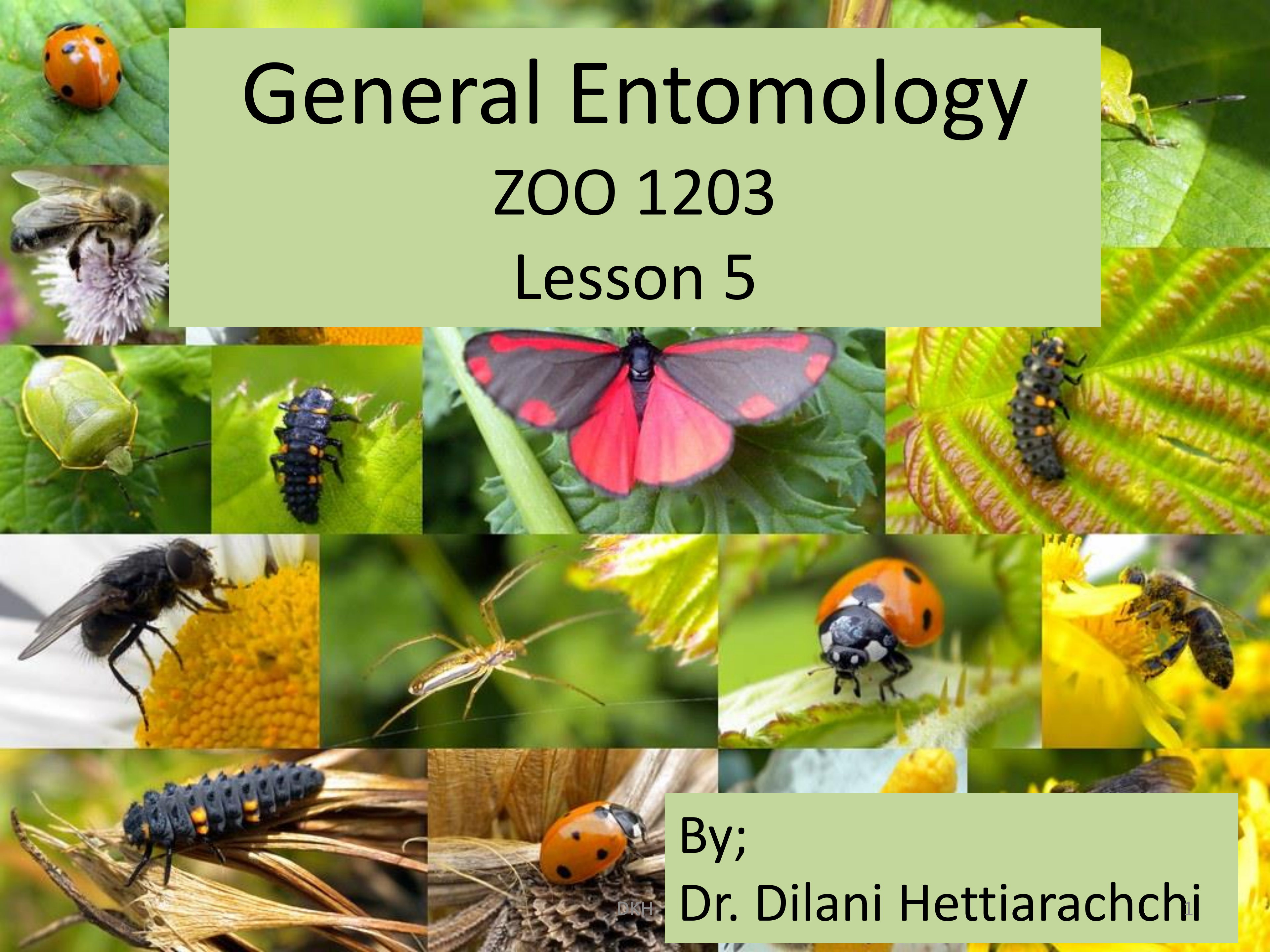


General Entomology

ZOO 1203

Lesson 5

By;
Dr. Dilani Hettiarachchi

Lesson Objectives

After completion you should be able to;

- Describe the following systems;
  - Digestive system
  - Circulatory system

Digestive system

- Varying food intake – watery xylem sap, blood, dry wood, bacteria and algae, internal tissues of other insects, leaf, stem, etc.

- Mouthparts are diverse correlate with the diet

- Sometimes larvae might have different feeding habits than the adults

- Therefore their mouth parts also changes and there are differences in the digestive tracts too

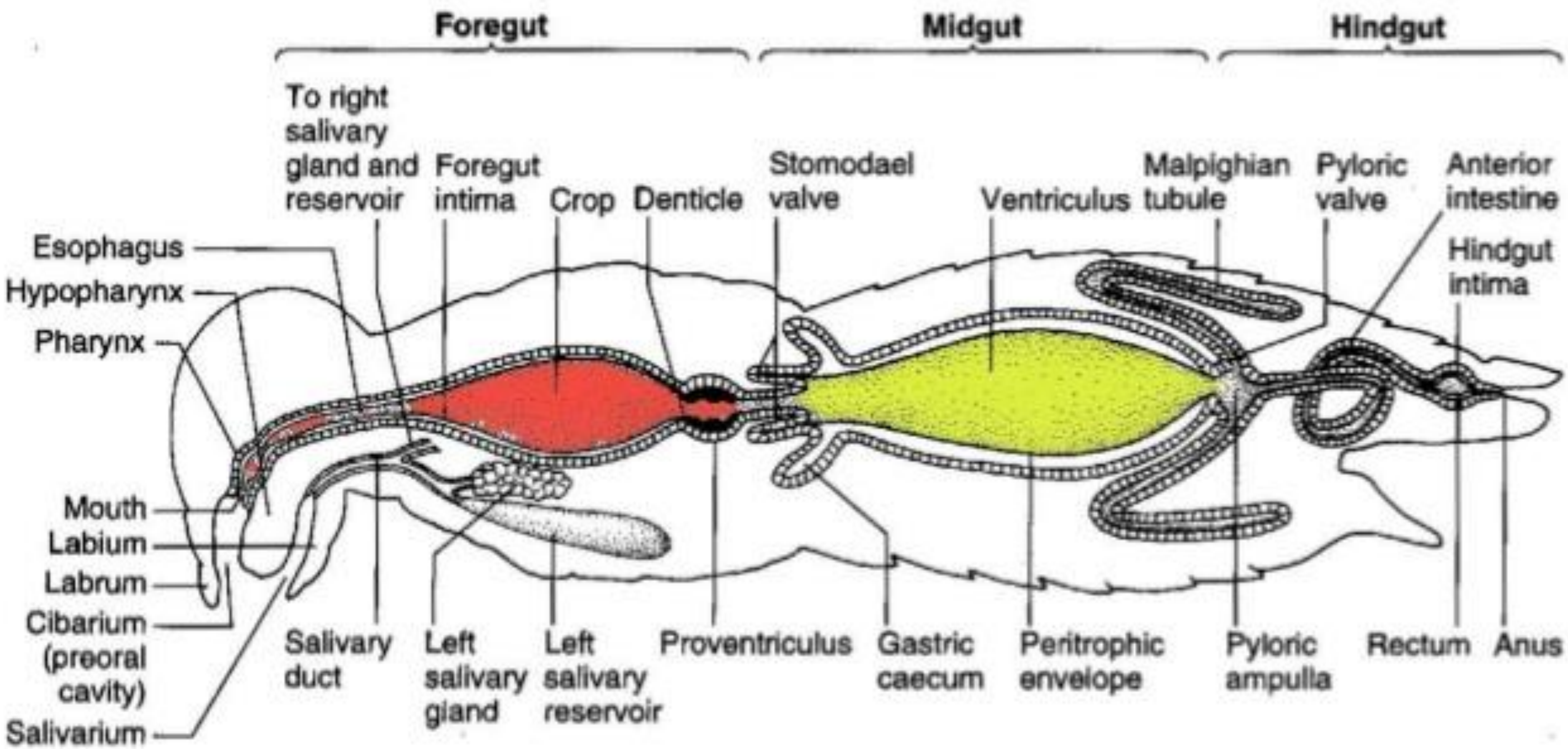
Structure of the gut

- It is a tube that extend from mouth to anus
- It is differentiated into 3 parts;

  1. fore gut
     - concerned with ingestion, storage, grinding and transport

  2. mid gut
     - digestive enzymes are produced, secreted and absorption of the products of digestion occur

  3. hind gut
     - remaining material in the gut lumen together with urine coming from malpighian tube enter
     - absorption of water, salt and other essentials occur
     - eliminate


Foregut
Midgut
Hindgut
To right salivary gland and reservoir
Foregut intima
Crop
Denticle
Stomodael valve
Ventriculus
Malpighian tubule
Pyloric valve
Anterior intestine
Hindgut intima
Esophagus
Hypopharynx
Pharynx
Mouth
Labium
Labrum
Cibarium (preoral cavity)
Salivarium
Salivary duct
Left salivary gland
Left salivary reservoir
Proventriculus
Gastric caecum
Peritrophic envelope
Pyloric ampulla
Rectum
Anus

- Gut epithelium is one cell layer thick

- On top of a basement membrane which is just above the muscle layer

- Fore gut and hind gut have a cuticular lining but mid gut does not

- Fore gut;
    - subdivided into pharynx, oesophagous and a crop (storage area)
    - insects that ingest solid food usually have a grinding organ called a proventriculus or a gizzard
    - proventriculus is found in orders Orthoptera, Blattodea, Mantodea, Phasmatodea, Dermeptera

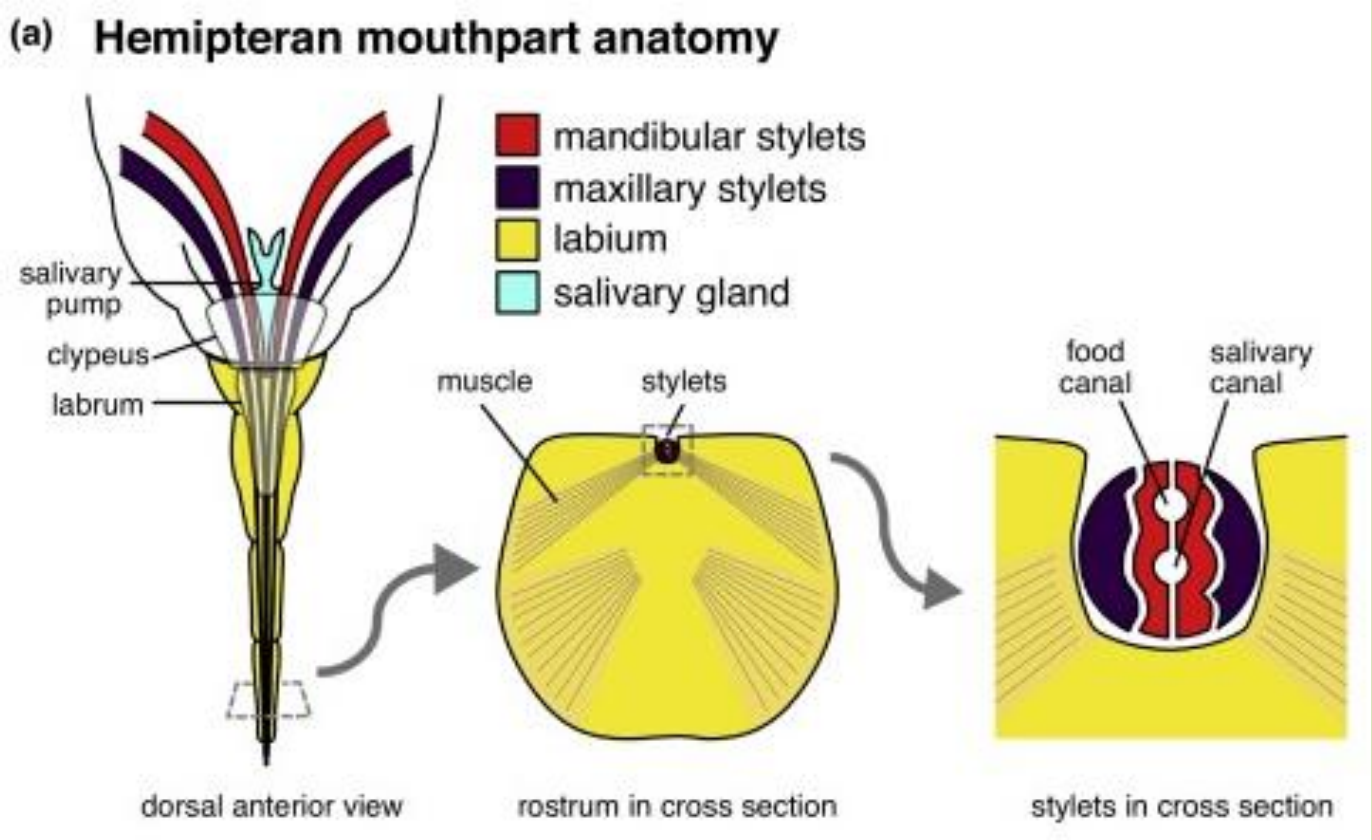
- Salivary glands are situated in the mouth

- Salivary secretion dilute the ingested food and adjust the pH

- Many Hemipterans (bugs) → produce two types of saliva

- Lepidoptera → labial glands produce silk, mandibular glands produce saliva

- Thus several types of secretory cells may occur in salivary glands of one insect



- Crop is capable of digesting the food

- There is a sphincter that control the flow of the food to the mid gut

Mid gut

- two main areas;
1. tubular ventriculus
2. blind-ending diverticula called caeca

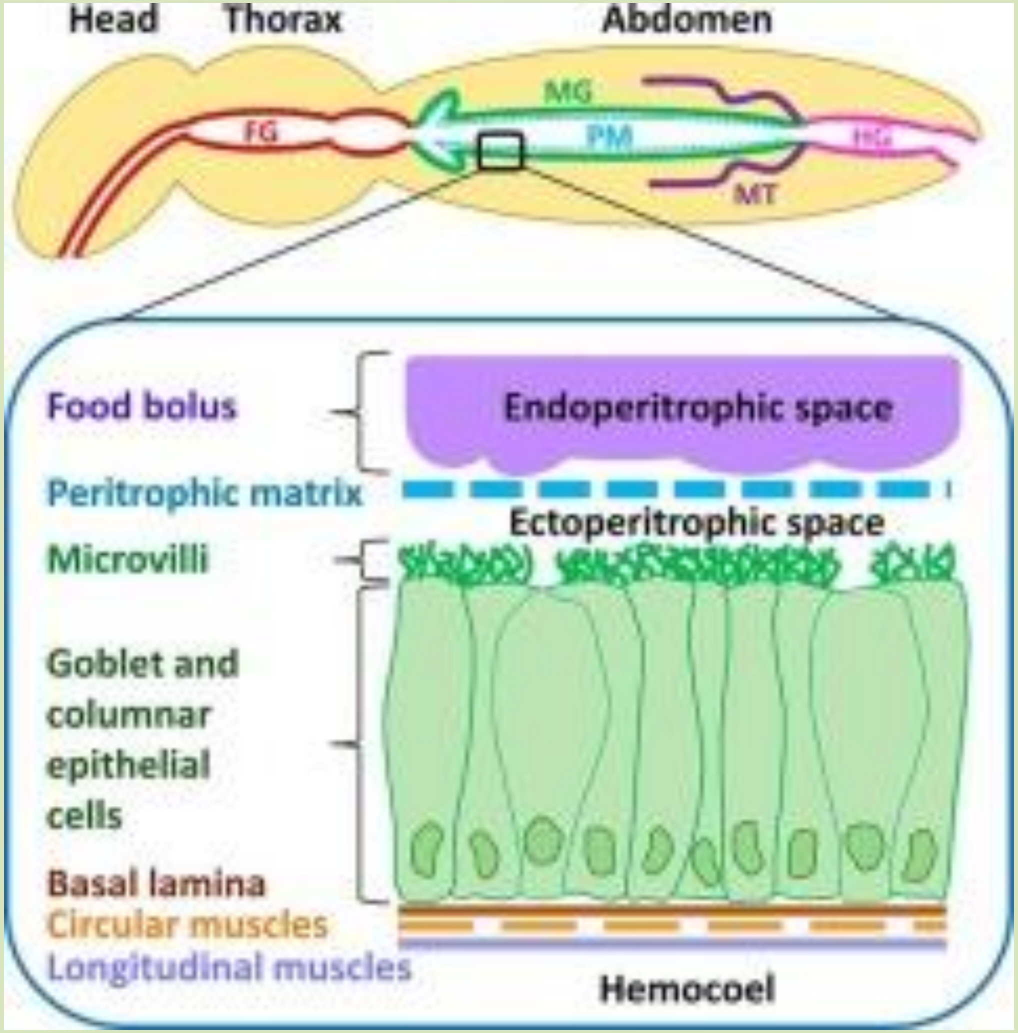
- most cells of the mid gut are structurally similar

- they are columnar cells with microvilli

- mid gut of most insects are separated from the food by a thin membrane named **peritrophic membrane** (Hemipterans do not have this)

- This membrane consists of network of chitin fibrils in a protein - carbohydrate matrix

- It protects the epithelium against food abrasion and microorganisms and has other functions based on compartmentalization of enzymes.



Hind gut

- contain the entry point of malpighian tubules
- it is connecting to a pylorus
- pyloric sphincter is present
- this is followed by the ileum, colon and rectum
- main function of the hind gut is absorption of water, salt and other useful substances from the faeces and urine

Digestion of food

- Digestion involves breaking large molecules into small monomers via the action of series of enzymes

- Digestion starts when it touches the saliva as it contain enzymes

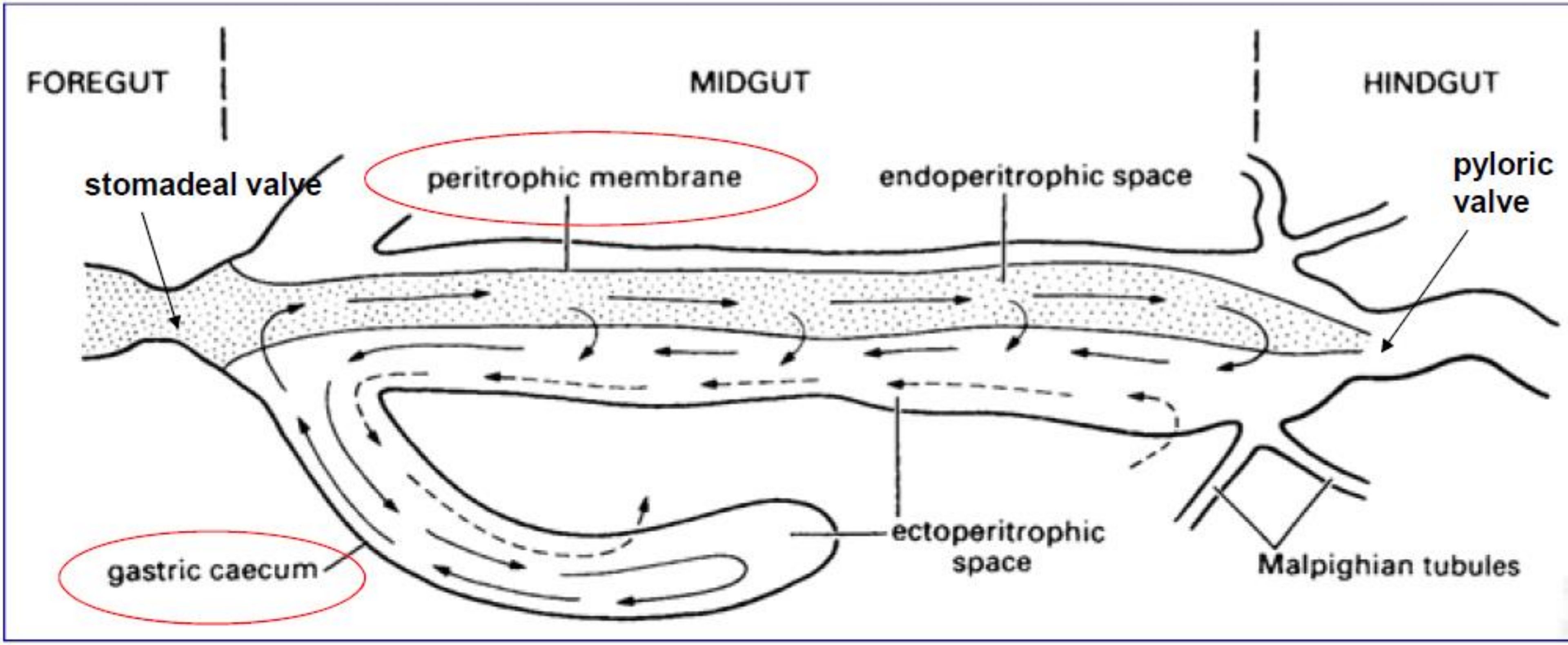
- This digestion can occur externally before the food is ingested
    - When injected into the host tissue (leafhopper, homoptera)
    - Or exuded over the food (house fly, Diptera)
    - Much of the digestion occur externally in extreme cases (diving beetles, Coleoptera)

- Slightly digested food is passed to the crop

- Then it is moved to the proventiculus – grinding

- In the mid gut many digestive enzymes comes into contact with the food

- Peritrophic membrane is in the mid gut
  - It contain sieves/ pores
  - Allows the passage of small molecules
  - Restrict entry of large molecules, bacteria and food particles directly accessing the midgut cells
  - In some insects most of the digestion take place inside this membrane called the endoperitrophic space



- Smaller particles will diffuse out from the endoperitrophic space to the ectoperitrophic space
- Final phase of digestion occurs in the surface of the mid gut microvilli
- Thus this membrane helps to compartmentalize the digestion

- Insects have most of the major enzymes found in other animals

- There are; proteases, lipases and carbohydrases

  Eg: amylase, maltase, lipase, invertase, peptidases, etc.

- Blood-feeding species primarily have proteolytic enzymes

- Wood boring insects may have cellulase

- In the hind gut; undigested food are eliminated via anus

- Reabsorption of water, salt and essentials take place in the rectum – some have tracheal gills in rectum (larval form)

Fat body

- Larvae of holometabolous insects (having complete metamorphosis) and some adults have a fat boy tissue

- White or yellow tissue, loose sheets lying in the haemocoel

- Structure is not define well and taxonomically variable

- Often caterpillars and other larvae have a peripheral layer of fat body beneath the cuticle and a central layer around the gut

- Functions;

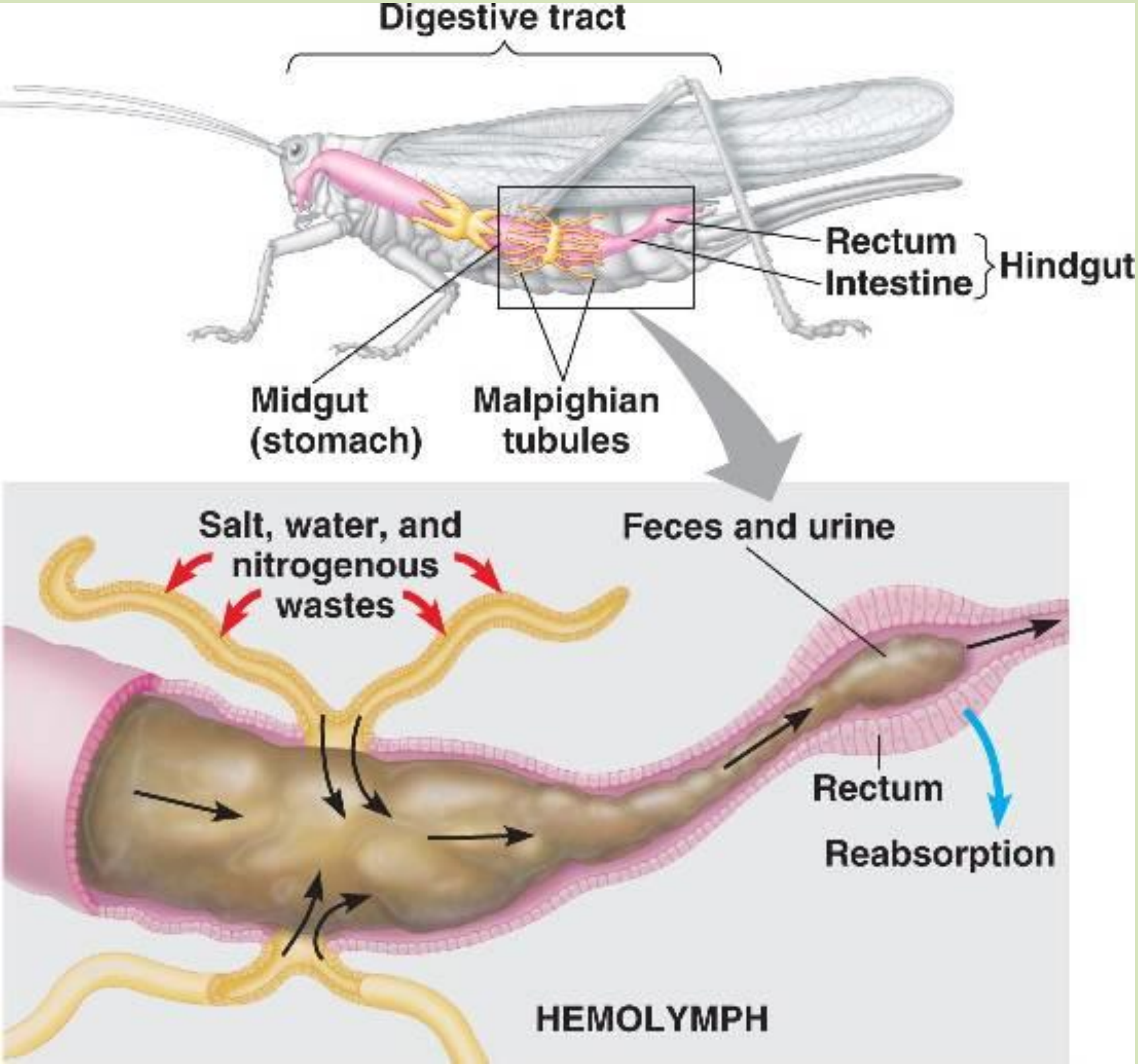
1. Metabolism of carbohydrates, lipids and nitrogenous compounds
2. Storage of glycogen, fat and protein
3. Synthesis and regulation of blood sugar
4. Synthesis of major haemolymph proteins such as haemoglobin, vitellogenins for yolk formation

- Fat body can change activities in response to nutritional and hormonal signals

- Supply the insect growth, metamorphosis and reproduction

Excretory system

- Excretion → removal of waste products from the body
- Insect faeces are either in liquid form or as pallets
- Aquatic insects excrete diluted waste where as terestrial insects excrete concentrated waste to conserve water
- Insects have Malpighian tubules → outgrowth of the alimentary canal
- Malpighian tubules vary in number from 2 – 200 or more
- Free waving around the haemolymph and closed at the tips


Digestive tract
Rectum
Intestine
Hindgut
Midgut (stomach)
Malpighian tubules
Salt, water, and nitrogenous wastes
Feces and urine
Rectum
Reabsorption
HEMOLYMPH

- Malpighian tubules collect waste from the blood and concentrate them

- The concentrated waste is then added to the undigested material which gets released through the anus

- Most of the nitrogen is taken up in the form of uric acid salt (formed by the fat body)

- Amino acids, various ions, and water are absorbed

- Pottassium ion gradient is use to flow the urine to hind gut

- Hindgut water reabsorbed with other necessary things

Insect Nutrition

- ❖ Insects are heterotrophs
- ❖ A balance diet must include an assortment of essential components in sufficient quantities
- ❖ Major dietary components are as follows:

1. Carbohydrates

- Complex polysaccharides like starch and glycogen are broken down in to simple sugars by digestive enzymes
- They can be use to construct chitin
- does not have digestive enzymes to break structural proteins such as chitin and cellulose
- Insect feed on wood (eg: termites) rely in digestive enzymes secreted by symbiotic bacteria and/or protozoans living inside the digestive tract

2. Proteins

(Actin, Myosin, Resilin, Arthropodin, Enzymes, Hormones, Ribosomes, Peptides, Amino acids)

- Protease enzymes break down proteins into amino acids
- Use them in building or replacing enzymes, hormones, muscles, egg yolk, cuticle and many more
- Can be converted to carbohydrates
- This produce toxic waste products that are converted to uric acid
- It’s collected in the malphighian tubules

- Insects and humans require the same ten essential amino acids

Lysine, tryptophan, histidine, phenylalanine, leucine, isoleucine, threonine, methionine, valine, and arginine

3. Lipids

(Fats (triglysarides): oils, waxes, resins; Fatty acids: cell membranes, pheramones; Steroids: Hormones, Cholestrol)

- Serve primarily as energy storage molecules

- Broke down to glyserol and fatty acids by lipase in the mid gut

- Fatty acid further digested to acetyl by thiokinase enzyme in the mitochondria

- With the co-enzyme A it will process through the Kreb's cycle to yield energy

- Insects does not have a mechanism to synthesis steroids

- Thus must obtain steroids (especially cholesterol) from diet

- Steroids use to make hormones (eg: ecdysteroids) and growth factors

- Insects feeding on steroid-deficient diets generally survive as immature but fail to molt properly into the adult stage

4. Nucleis Acid

(DNA, RNA; Purines and Pyrimidines, Mono and Di nucleotides: ATM, Cyclic AMP, NAD, NADP and FAD)

Sugars (ribose and deoxyribose)
nucleotides (adinine, guanine, cytosine, urasil and thyamine)
phosphate

- Individual cells reuse these components for synthesis of new DNA and RNA

- Dinucleotides such as NAD, NADP, FAD are important oxidation-reduction co-enzymes

- ATP – energy currency: Cyclic AMP – second messenger

- In addition:
  - water
  - vitamins
  - minerals

- Most terrestrial insects are highly adapted for water conservation – get most of the water from their food

- Some can live for months only by metabolic water (protein synthesis/ chitin synthesis)

- Vitamins include a number of complex organic molecules

- Humans must have a dietary source of both fat-soluble and water soluble vitamins

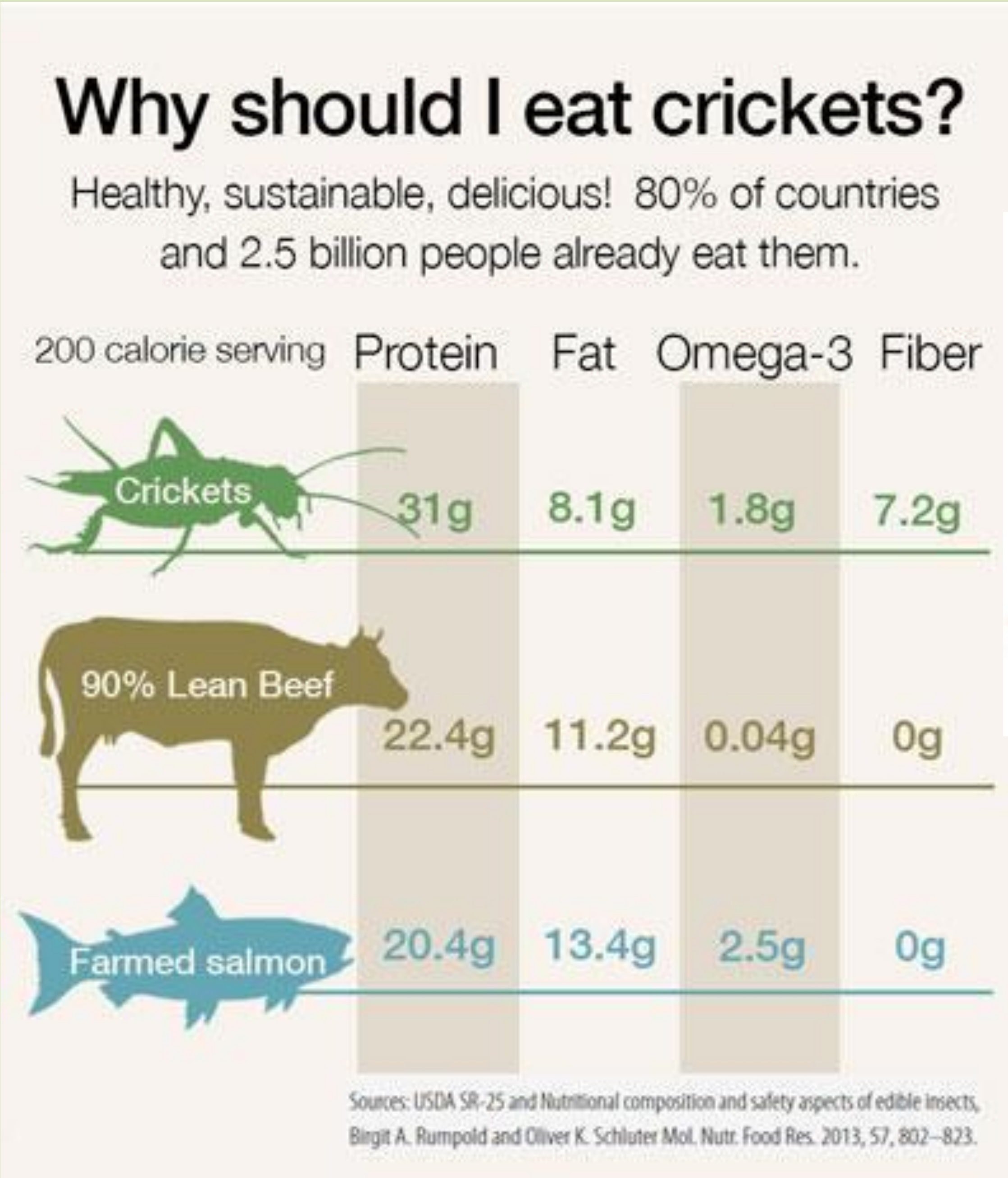
- Insects have the ability to synthesize their own fat-soluble vitamins (A,D,E and K) from other compounds

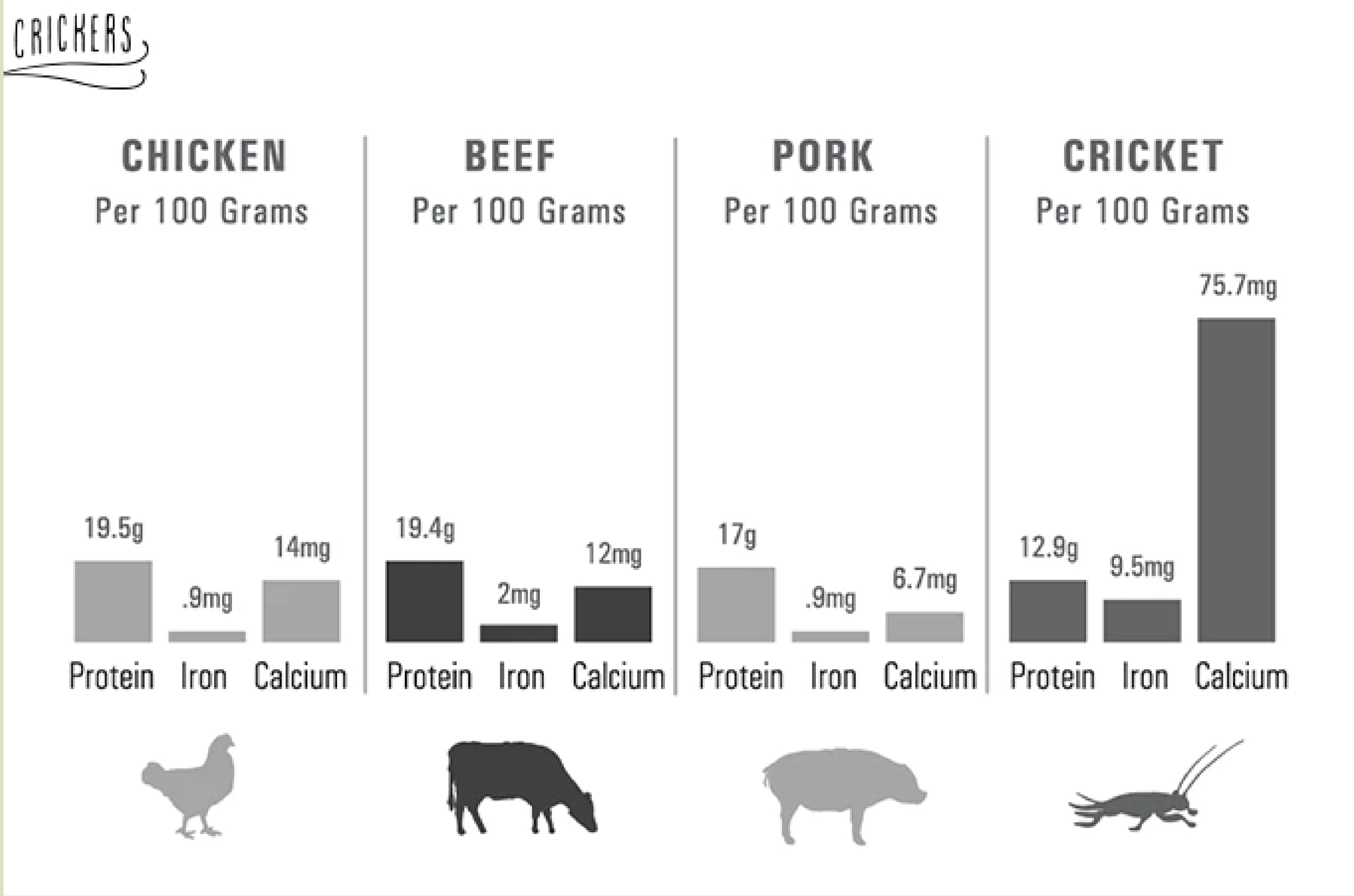
- But they still need a dietary source for most water-soluble vitamins such as Thiamine (B1), riboflavin (B2), Pyridoxine (B6), nicotinic acid, ascorbic acid, pantothenic acid and biotin

- Minerals are need in small amounts

- Iron and copper – needed for cytochromes
- Calcium, sodium and potassium – nerves and muscles
- Phosphorus – cell membrane
- Sulfur – component of some proteins, sclerotization of exoskeleton

Insect as human food




CRICKERS
CHICKEN
Per 100 Grams
19.5g
.9mg
14mg
Protein
Iron
Calcium
BEEF
Per 100 Grams
19.4g
2mg
12mg
Protein
Iron
Calcium
PORK
Per 100 Grams
17g
.9mg
6.7mg
Protein
Iron
Calcium
CRICKET
Per 100 Grams
12.9g
9.5mg
75.7mg
Protein
Iron
Calcium

Circulatory system

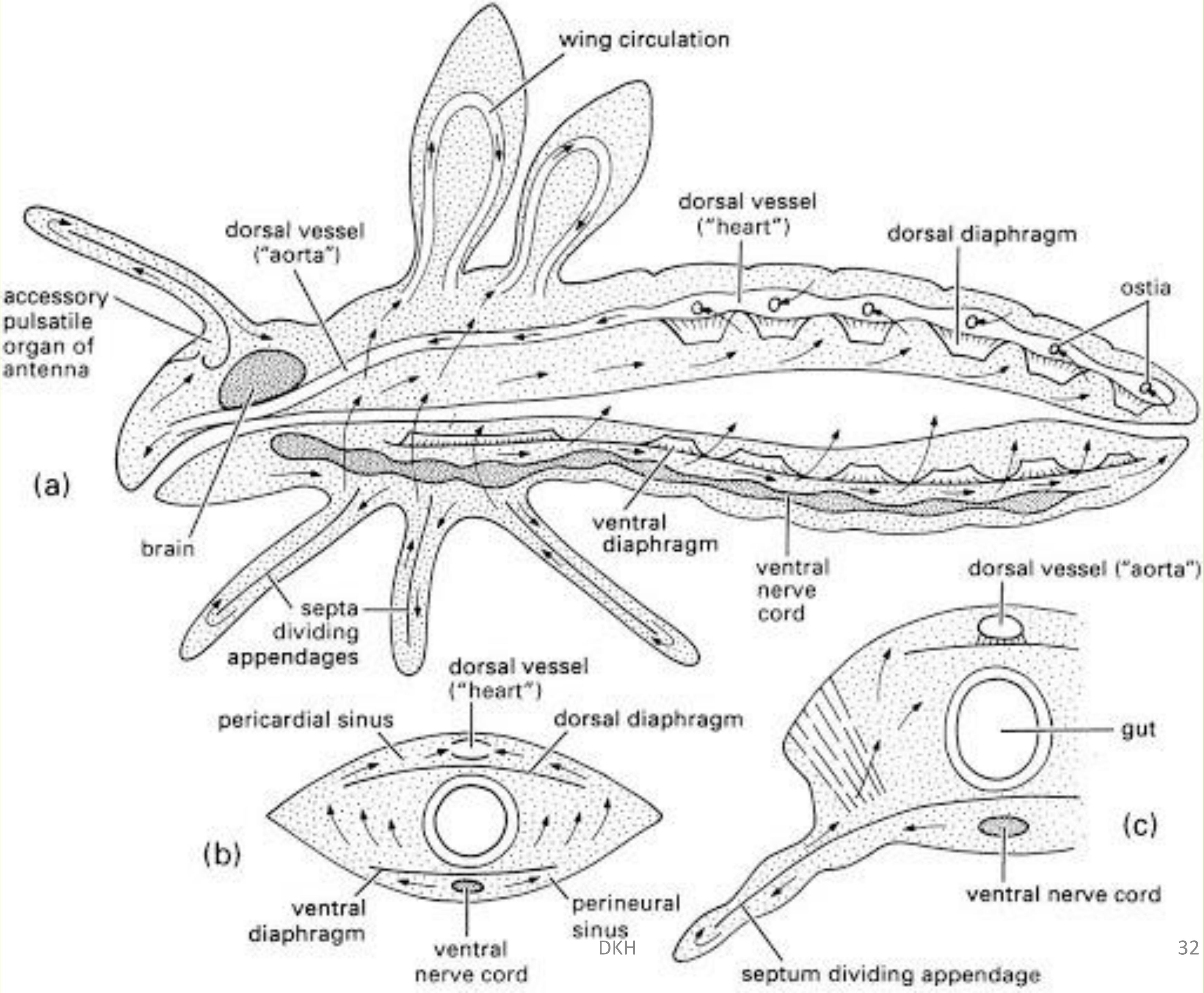
- Have an open circulatory system that is not restricted to vessels

- Blood flow through the open body cavity, the haemocoel and supply the organs and tissues

- Main part is the dorsal vessel → two parts; heart at the rear and the aorta in the front

- Heart is a chambered organ with an opening called ostia

- Draw blood through the ostia and pump forward to the aorta


wing circulation
dorsal vessel ("aorta")
accessory pulsatile organ of antenna
dorsal vessel ("heart")
dorsal diaphragm
ostia
(a)
brain
ventral diaphragm
ventral nerve cord
septa dividing appendages
dorsal vessel ("aorta")
gut
(c)
ventral nerve cord
septum dividing appendage
dorsal vessel ("heart")
pericardial sinus
dorsal diaphragm
(b)
ventral diaphragm
ventral nerve cord
perineural sinus

- Aorta is a simple tube that carries blood to the head capsule
- Below the heart have wing-shaped muscles know as alary muscles → partition between the heart and the body cavity
- This complete partition is known as **dorsal diaphragm**
- Chamber occupying the heart is named **dorsal sinus**
- In many insects there is a **ventral diaphragm** – guides the blood flow back and to the sides – lies in the floor of the body cavity

- Blood of an insect is called a haemolymph – watery fluid
- Main function of blood in insects is to transport nutrients, wastes and hormones
- Circulation of the haemolymph is aided by the peristaltic contractions of the ventral diaphragm and the expansion of the abdomen
- Circulation to appendages is unidirectional by various septa, tubes, valves and pumps
- There are accessary pulsating organs (“auxiliary hearts”) aid in circulation

- Accessary pulsating organs ("auxiliary hearts") deliver blood to appendages
- They are present at the base of antennae, at the base of wings, and sometimes the legs
- Insect circulatory system shows an impressive degree of coordination between the activities of the dorsal vessel, diaphragm and accessary pumps
- This is mediated by both nervous and hormonal regulation
- Oxygen is delivered by the tracheal system not the circulatory system (no red blood cells)

Respiration system

- Respiration involves the intake, transport and utilization of oxygen by cells and tissues and the removal of carbon dioxide from the body

- Insects respire by diffusion of atmospheric gases directly across membranes into the cells

- And the tracheal system – main system for transport of gasses in insects

Tracheal System

- Composed of series of branching tubes called tracheae
- They open to the outside by an opening; spiracles
- Spiracles are on each side of most body segments
- Sometimes these spiracles have openings in which the opening can be temporarily closed
- Spiracles open to tracheal tubules which will be connected to large trachea (tracheal trunk) that runs along the body

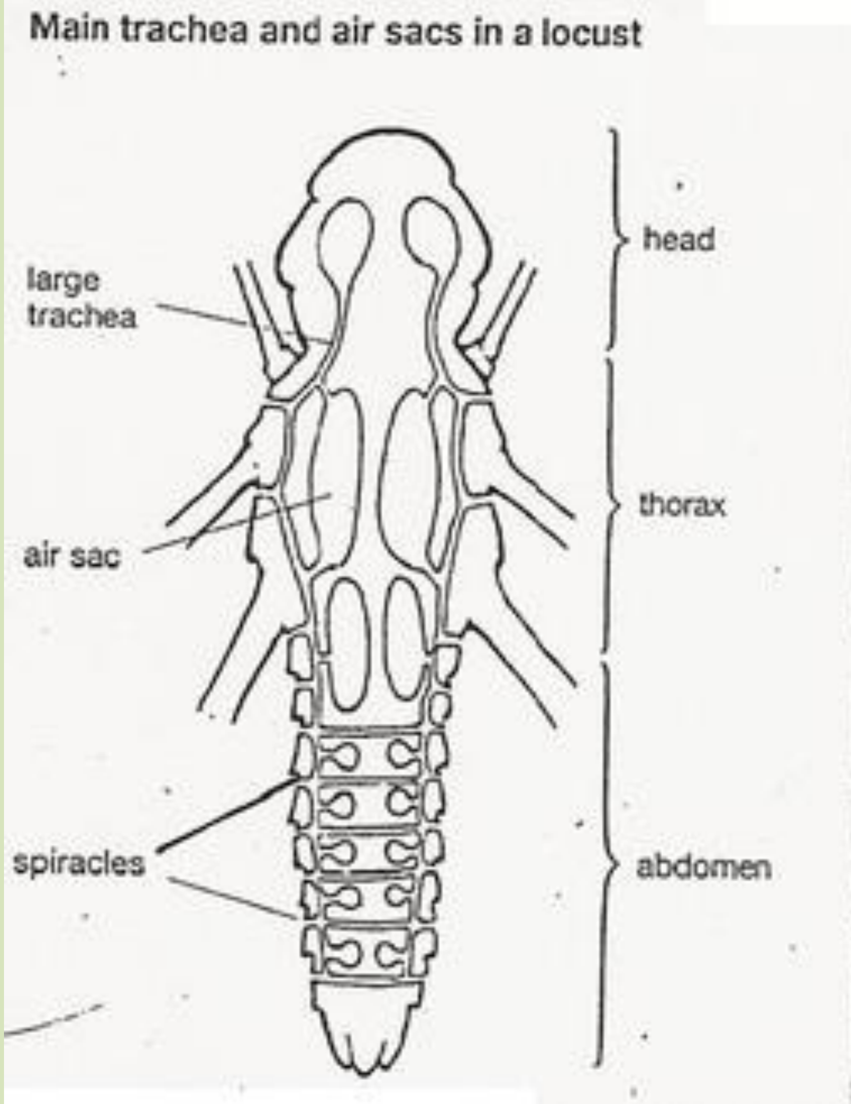
- In fast-flying insects such as the house fly, these tracheal trunks have enlargements called air sacs → increase ventilation

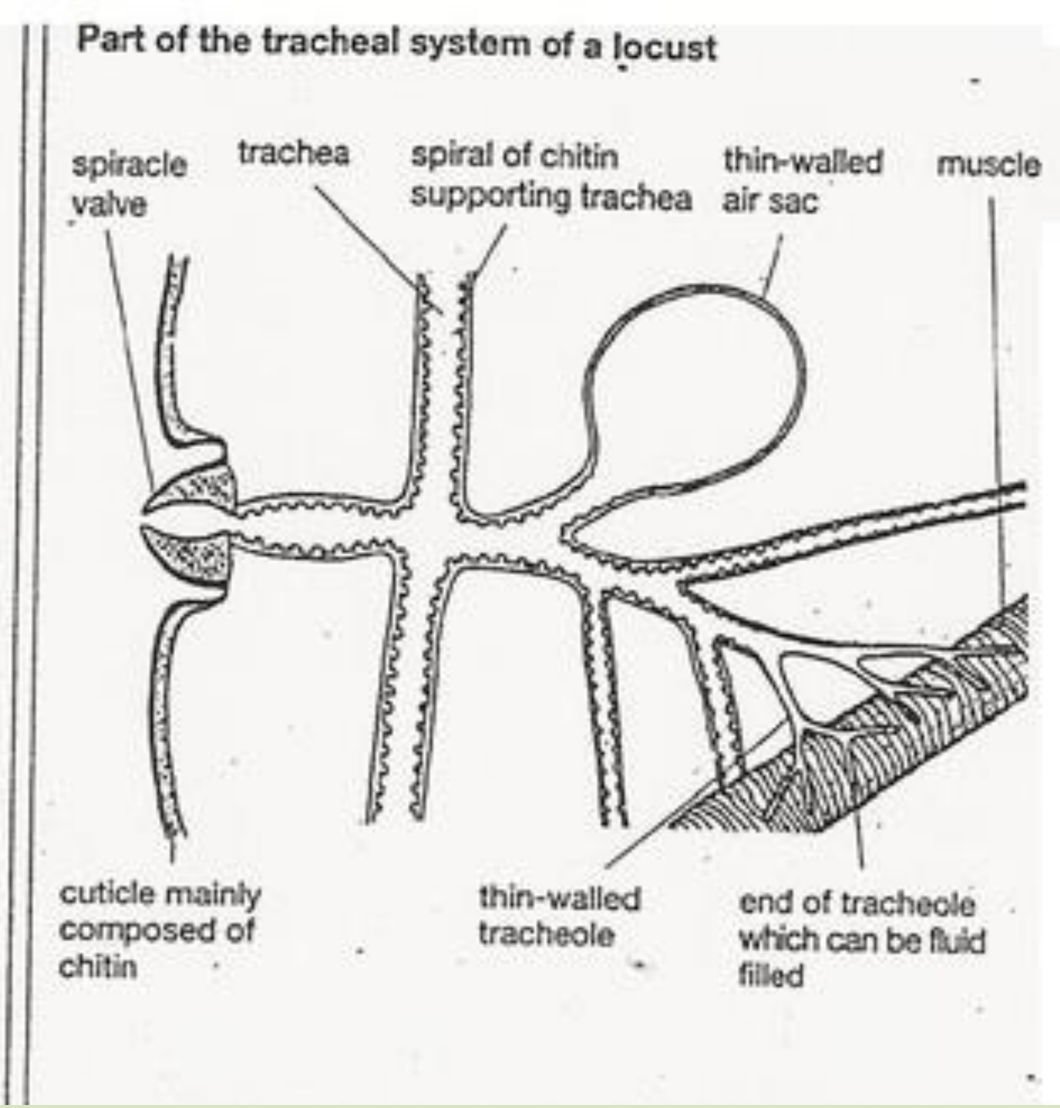
- Branches arise from the tracheal trunk in each segment and become increasingly fine and branched

- At the end these will be divided into minute tracheoles and penetrate deeply into muscle fibers

- Spiracles open to the outside thus called an open system

- Some have close systems where the spiracles become non functional


Main trachea and air sacs in a locust
large trachea
air sac
spiracles
head
thorax
abdomen


Part of the tracheal system of a locust
spiracle valve
trachea
spiral of chitin supporting trachea
thin-walled air sac
muscle
cuticle mainly composed of chitin
thin-walled tracheole
end of tracheole which can be fluid filled

- Many aquatic immature have the closed system (eg: mayfly, stone fly)

- Instead of spiracles they have a network of tracheoles that runs under the skin or into external gills (in Odonata/ dragonfly nymph)

- There are some other forms of respiratory organs such as air stores in aquatic insects, air tubes used like a snorkel to siphon air from the surface (ex: immature mosquitoes and water scorpion)

Respiration in aquatic insects

- Variety of adaptations

1. Cuticular respiration

- Relatively thin integument

- Permeable to oxygen and carbon dioxide

- Mostly small inactive animals – especially those living in cold, fast-moving streams

- Larger insects – may need other adaptations

2. Gills

- Usually outgrowths of the tracheal system

- Covered with thin layer of cuticle that is permeable to both oxygen and carbon dioxide

- Mayfly and damselfly – leaf-like and located on the sides or rare of the abdomen

- Fanning movement of the gills keep them contact with fresh water

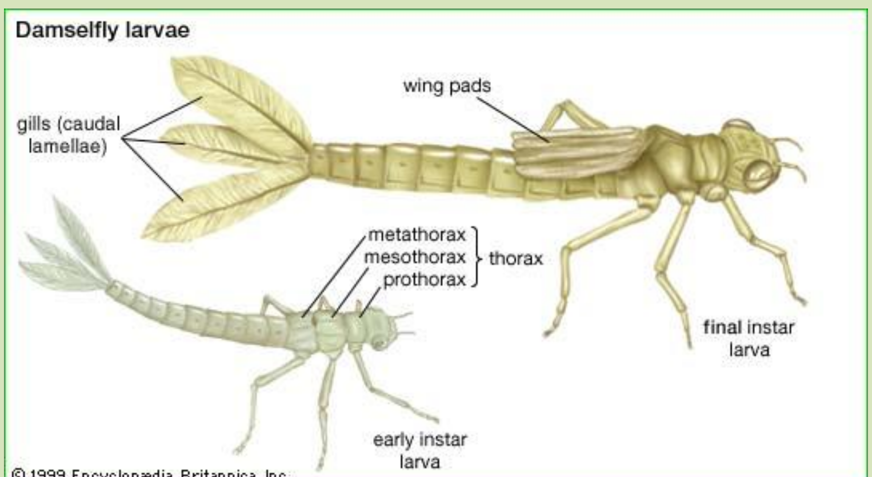
- Stoneflies – filamentous gills on the thorax and abdomen

- Dragonfly – internal gills associated with a thorax or abdomen

- Water is circulated in and out of the anus by muscular contraction of the abdomen

- They show jet propulsion

- Used when need to escape from predators



3. Breathing tubes/ siphon

- Although many aquatic insects live under water they take air from the surface

- It works like a snorkel

- Eg: mosquito larvae, water scorpion, rat-tailed maggots

- Opening at the end of the siphon is guarded by a ring of closely spaced hairs with a waterproofing coating

- When insect dive water pressure pushes the hairs close together so they seal off the opening

4. Air bubbles

- Some carry a bubble of air with them whenever they dive
- Bubble maybe held under the elytra or it may be trapped against the body by specialized hairs
- Bubble usually cover one or more spiracles
- Provide only a short term supply of oxygen
- Bubble will also collect some oxygen from the water through passive diffusion
- Larger the bubble more it can stay

5. Plastrons

- Series of closely-spaced hydrophobic hairs (setae)

- It create an “airspace” next to the body

- Air trapped within a plastron operates as a physical gill/air bubble

- Insects that remain permanently submerged (ex: riffle beetle, family Elmidae) or lack the ability to reach the surface (ex: eggs of mosquitoes) have plastrons

- Visible underwater as a thin, silvery film

6. Hemoglobin

- Respiratory pigment

- It occurs rarely in insects

- Found in certain midges (family Chironomidae) known as bloodworms

- Back up supply – last only few minutes but enough for the insect to move into oxygenate water


THANK YOU!